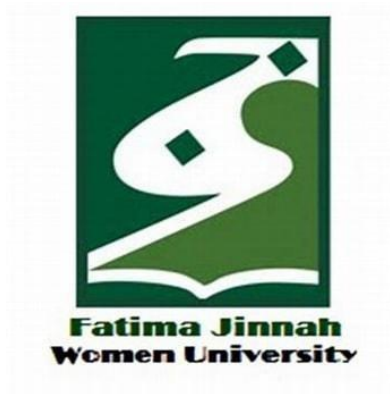


CLOUD COMPUTING

LAB NO# 07



SUBMITTED TO:

Sir Shohaib

SUBMITTED BY:

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Reg No. 2023-BSE-007

Semester: V-A

CLOUD COMPUTING

LAB NO# 07

Task 1 — Print & Filter Environment Variables

Step 1: Print all environment variables

Run this command:

Printenv

```

XDG_SEAT=seat0
PWD=/home/amina
LOGNAME=amina
XDG_SESSION_TYPE=ttty
SYSTEMD_EXEC_PID=1372
HOME=/home/amina
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01:34:ln=01:36:mh=00:pi=40:33:so=01:35:do=01:35:bd=40:33:01:cd=40:33:01:or=40:31:01:mi=00:sw=37:41:sg=
x=01:32:*.tar=01:31:*.tgz=01:31:*.arc=01:31:*.arj=01:31:*.taz=01:31:*.lha=01:31:*.lz4=01:31:*.lzh=01:31:*.lzm=01:31:*.t
1:31:*.zip=01:31:*.z=01:31:*.gz=01:31:*.lrz=01:31:*.lz=01:31:*.lzo=01:31:*.xz=01:31:*.zst=01:31:*.tzt=01:31:
2=01:31:*.t=01:31:*.deb=01:31:*.rpm=01:31:*.jar=01:31:*.war=01:31:*.ear=01:31:*.sar=01:31:*.rar=01:31:*.alz=01:31:*.ace
31:*.rz=01:31:*.cab=01:31:*.wim=01:31:*.sum=01:31:*.dwm=01:31:*.esd=01:31:*.avif=01:35:*.jpg=01:35:*.jpeg=01:35:*.mjpg=0
35:*.pnm=01:35:*.pgm=01:35:*.ppm=01:35:*.tga=01:35:*.xbm=01:35:*.xpm=01:35:*.tif=01:35:*.tiff=01:35:*.png=01:35:*.svg=0
3:*.mov=01:35:*.mpg=01:35:*.mpeg=01:35:*.m2v=01:35:*.mkv=01:35:*.webm=01:35:*.webp=01:35:*.ogm=01:35:*.mp4=01:35:*.m4v=01
*.nuv=01:35:*.wmv=01:35:*.asf=01:35:*.rm=01:35:*.rmvb=01:35:*.flc=01:35:*.avi=01:35:*.fli=01:35:*.flv=01:35:*.gl=01:35:*.
01:35:*.cgm=01:35:*.emf=01:35:*.ogv=01:35:*.ogx=01:35:*.aac=00:36:*.au=00:36:*.flac=00:36:*.m4a=00:36:*.mid=00:36:*.midi
36:*.ogg=00:36:*.ra=00:36:*.wav=00:36:*.oga=00:36:*.opus=00:36:*.spx=00:36:*.xspf=00:36:*.x=00:90:*.#*=00:90:*.bak=00:90:*.
g-new=00:90:*.dpgk-old=00:90:*.dpgk-tmp=00:90:*.old=00:90:*.orig=00:90:*.part=00:90:*.rej=00:90:*.rpmnew=00:90:*.rpmorig
=00:90:*.ucf-dist=00:90:*.ucf-new=00:90:*.ucf-old=00:90:
MEMORY_PRESSURE_WATCH=/sys/fs/cgroup/system.slice/system-getty.slice/getty@tty1.service/memory.pressure
INVOCATION_ID=0c94d2e18a9845c39b6833d5387b4299
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=linux
LESSOPEN=| /usr/bin/lesspipe %s
USER=amina
SHLVL=1
XDG_VTNR=1
XDG_SESSION_ID=2
XDG_RUNTIME_DIR=/run/user/1000
XDG_DATA_DIRS=/usr/share/gnome:/usr/local/share:/usr/share:/var/lib/napd/desktop
HUSHLOGIN=FALSE
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
MAIL=/var/mail/amina
_=/usr/bin/printenv
amina@0203-bse-007:~$

```

Step 2: Filter specific variables (SHELL, HOME, USER)

Run **all three commands together**, one after another (do **not** take separate screenshots):

```
printenv | grep SHELL
```

```
printenv | grep HOME
```

```
printenv | grep USER
```

```
amina@2023-bse-007:~$ printenv | grep SHELL
SHELL=/bin/bash
amina@2023-bse-007:~$ printenv | grep HOME
HOME=/home/amina
amina@2023-bse-007:~$ printenv | grep USER
USER=amina
amina@2023-bse-007:~$
```

Task 2: Export DB_* Variables Temporarily & Observe Scope

Step 1: Define all DB_* variables (temporary)

Run **all three export commands one after another** in the same terminal:

```
export DB_URL="postgres://db.example.local:5432/mydb"
```

```
export DB_USER="labuser"
```

```
export DB_PASSWORD="labpass123"
```

```
amina@2023-bse-007:~$ export DB_URL="postgres://db.example.local:5432/mydb"
amina@2023-bse-007:~$ export DB_USER="labuser"
amina@2023-bse-007:~$ export DB_PASSWORD="labpass123"
amina@2023-bse-007:~$
```

Step 2: Echo all three variables

Run these **together**, in order:

```
echo "$DB_URL"
```

```
echo "$DB_USER"
```

```
echo "$DB_PASSWORD"
```

```
amina@2023-bse-007:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
amina@2023-bse-007:~$ echo "$DB_USER"
labuser
amina@2023-bse-007:~$ echo "$DB_PASSWORD"
labpass123
amina@2023-bse-007:~$
```

Step 3: Show all DB_* variables using grep

Run **one command only**:

```
printenv | grep '^DB_'
```

```
amina@2023-bse-007:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
amina@2023-bse-007:~$
```

Step 4: Close shell and verify variables are gone

1. Exit the current shell:

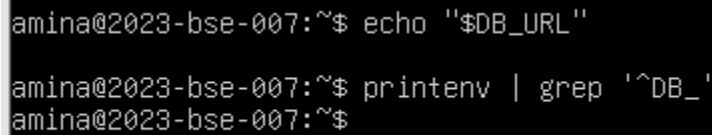
```
exit
```

2. Open a **new terminal / SSH session**

3. Run **both checks together**:

```
echo "$DB_URL"
```

```
printenv | grep '^DB_'
```



```
amina@2023-bse-007:~$ echo "$DB_URL"
amina@2023-bse-007:~$ printenv | grep '^DB_'
amina@2023-bse-007:~$
```

Task 3: Make DB_* Variables Persistent in ~/.bashrc

Step 1: Add DB_* variables to ~/.bashrc

Open the file in **vim**:

```
vim ~/.bashrc
```

Scroll to the **very end** of the file and add:

```
# Lab 7 persistent DB variables
```

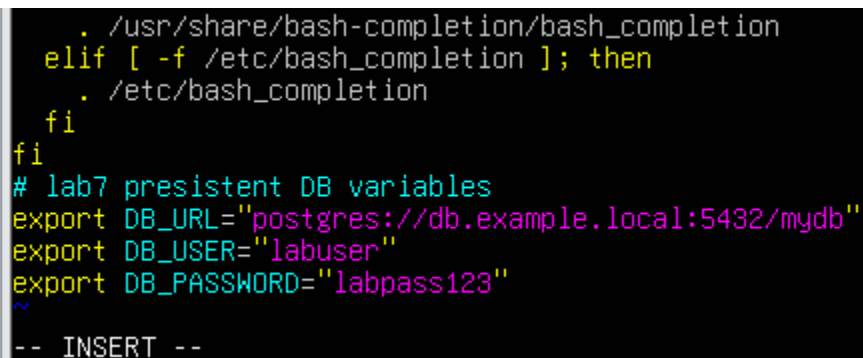
```
export DB_URL="postgres://db.example.local:5432/mydb"
```

```
export DB_USER="labuser"
```

```
export DB_PASSWORD="labpass123"
```

Save and exit:

Esc :wq Enter

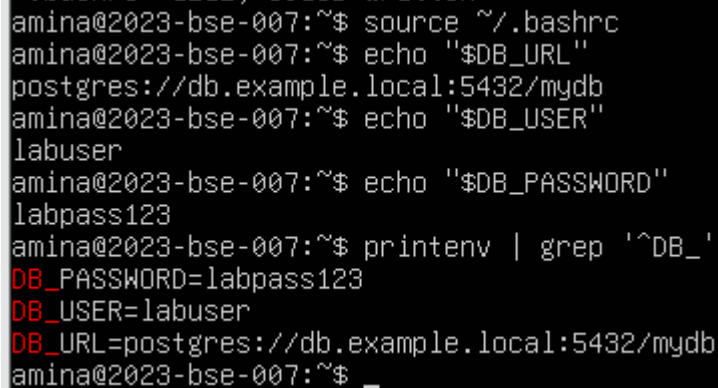


```
. /usr/share/bash-completion/bash_completion
elif [ -f /etc/bash_completion ]; then
. /etc/bash_completion
fi
# lab7 persistent DB variables
export DB_URL="postgres://db.example.local:5432/mydb"
export DB_USER="labuser"
export DB_PASSWORD="labpass123"
-- INSERT --
```

Step 2: Source ~/.bashrc and verify (grouped)

Run **all commands below together**, in this order, and capture **one single screenshot**:

```
source ~/.bashrc
echo "$DB_URL"
echo "$DB_USER"
echo "$DB_PASSWORD"
printenv | grep '^DB_'
```



```
amina@2023-bse-007:~$ source ~/.bashrc
amina@2023-bse-007:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
amina@2023-bse-007:~$ echo "$DB_USER"
labuser
amina@2023-bse-007:~$ echo "$DB_PASSWORD"
labpass123
amina@2023-bse-007:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
amina@2023-bse-007:~$ _
```

Step 3: Restart terminal and verify persistence

1. Close the terminal or SSH session
2. Open a **new terminal**
3. Run **both commands together**:

```
echo "$DB_URL"
printenv | grep '^DB_'
```



```
amina@2023-bse-007:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
amina@2023-bse-007:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
amina@2023-bse-007:~$ _
```

Task 4: System-wide Variable, Welcome Script, and PATH

Step 1: View /etc/environment (before change)

Run:

```
sudo cat /etc/environment
```

```
amina@2023-bse-007:~$ sudo cat /etc/environment
[sudo] password for amina:
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
amina@2023-bse-007:~$ _
```

Step 2: Show current PATH (before change)

Run:

echo "\$PATH"

```
amina@2023-bse-007:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
amina@2023-bse-007:~$ _
```

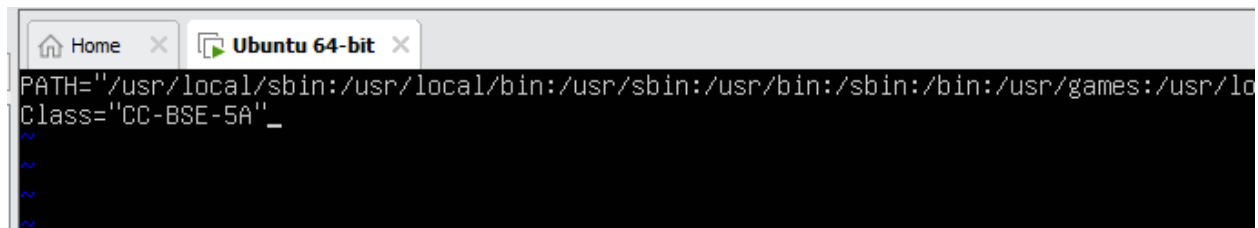
Step 3: Edit /etc/environment and add Class variable

Open file:

sudo vim /etc/environment

Add **at the end** (replace <your_class_name> appropriately, e.g. CC-BSCS-6A):

Class="CC-<your_class_name>"



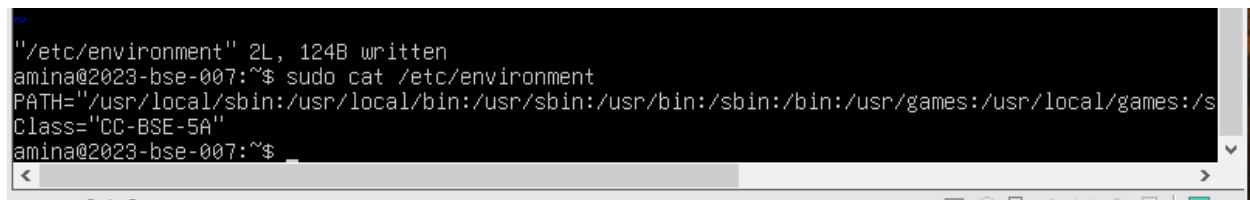
```
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-BSE-5A"
~
~
~
~
```

Save and exit:

Esc :wq Enter

2. Verify the file contents:

sudo cat /etc/environment



```
"/etc/environment" 2L, 124B written
amina@2023-bse-007:~$ sudo cat /etc/environment
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-BSE-5A"
amina@2023-bse-007:~$ _
```

Step 4: Re-login and show Class & PATH together (grouped)

Run **both commands together**:

echo \$Class

echo "\$PATH"

```
12 additional security updates can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
amina@2023-bse-007:~$ echo "$Class"  
CC-BSE-5A  
amina@2023-bse-007:~$ echo "$PATH"  
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin  
amina@2023-bse-007:~$ _
```

Step 5: Create welcome script and make executable (grouped)

Run **both actions together**:

```
cat > ~/welcome <<'EOF'
```

```
#!/bin/bash
```

```
echo "Welcome to Cloud Computing $USER"
```

```
EOF
```

```
chmod +x ~/welcome
```

```
amina@2023-bse-007:~$ cat > ~/welcome <<'EOF'  
> #!/bin/bash  
> echo "welcome to cloud computing $USER"  
> EOF  
amina@2023-bse-007:~$ chmod +x ~/welcome  
amina@2023-bse-007:~$ _
```

Step 6: Run script using ./welcome

Run:

```
cd ~
```

```
./welcome
```

```
amina@2023-bse-007:~$ cd ~  
amina@2023-bse-007:~$ ./welcome  
welcome to cloud computing amina  
amina@2023-bse-007:~$ _
```

Step 7: Add home directory to PATH in ~/.bashrc

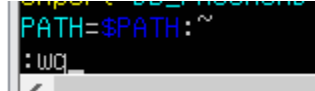
Open file:

```
vim ~/.bashrc
```

Add **ONLY this line at the end** (⚠ no export):

```
PATH=$PATH:~
```

Save and exit.

A terminal window showing the command `PATH=$PATH:~` being entered. The prompt is `~$` and the cursor is at the end of the line.

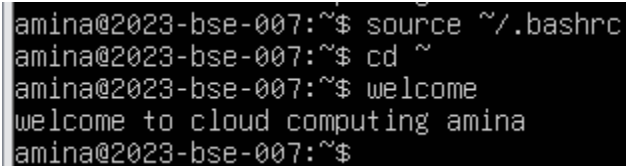
Step 8: Apply PATH change and run welcome without ./

Run **in this order**, capture together:

```
source ~/.bashrc
```

```
cd ~
```

```
welcome
```

A terminal window showing the execution of the commands `source ~/.bashrc`, `cd ~`, and `welcome`. The output of `welcome` is `welcome to cloud computing amina`. The prompt is `amina@2023-bse-007:~$`.

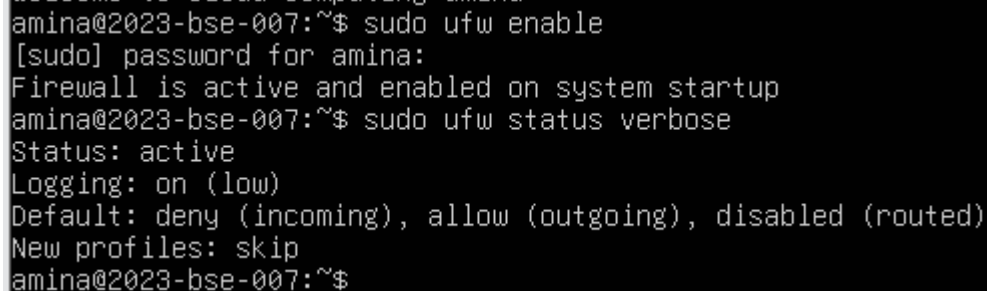
Task 5: Block and Allow SSH Using UFW

Step 1: Enable UFW and show status (grouped)

Run **both commands together**:

```
sudo ufw enable
```

```
sudo ufw status verbose
```

A terminal window showing the execution of the commands `sudo ufw enable` and `sudo ufw status verbose`. The output of `sudo ufw enable` is `[sudo] password for amina: Firewall is active and enabled on system startup`. The output of `sudo ufw status verbose` is `Status: active Logging: on (low) Default: deny (incoming), allow (outgoing), disabled (routed) New profiles: skip`. The prompt is `amina@2023-bse-007:~$`.

Step 2: Deny SSH (port 22) and show status (grouped)

Run **both commands together**:

```
sudo ufw deny 22/tcp
```

```
sudo ufw status numbered
```

```

amina@2023-bse-007:~$ sudo ufw deny 22/tcp
Rule added
Rule added (v6)
amina@2023-bse-007:~$ sudo ufw status numbered
Status: active

      To Action From
      --
[ 1] 22/tcp DENY IN Anywhere
[ 2] 22/tcp (v6) DENY IN Anywhere (v6)
amina@2023-bse-007:~$

```

Step 3: Attempt SSH from Windows host (should FAIL)

On your **Windows machine** (PowerShell or Command Prompt), run:

ssh username@<server_ip>

```

C:\Users\Lenovo>ssh amina@192.168.98.128
ssh: connect to host 192.168.98.128 port 22: Connection timed out
C:\Users\Lenovo>

```

Step 4: Allow SSH again, reload, and show status (grouped)

Back on the **VM (existing session)**, run **all commands together**:

sudo ufw allow 22/tcp

sudo ufw reload

sudo ufw status

```

amina@2023-bse-007:~$ sudo ufw allow 22/tcp
Rule updated
Rule updated (v6)
amina@2023-bse-007:~$ sudo ufw reload
Firewall reloaded
amina@2023-bse-007:~$ sudo ufw status
Status: active

      To Action From
      --
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
amina@2023-bse-007:~$

```

Step 5: SSH from Windows host again (should SUCCEED)

On **Windows host**, run again:

ssh username@<server_ip>

```
amina@2023-bse-007: ~
C:\Users\Lenovo>ssh amina@192.168.98.128
amina@192.168.98.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jan 10 02:17:57 PM UTC 2026

System load:  0.08               Processes:           226
Usage of /:   50.5% of 14.66GB   Users logged in:    1
Memory usage: 16%               IPv4 address for ens33: 192.168.98.128
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Fri Dec 26 08:04:10 2025 from 192.168.98.1
amina@2023-bse-007:~$
```

Task 6: Configure SSH Key-Based Login (Windows → Ubuntu)

On Windows Host (Client)

Step A1: Generate SSH key + list .ssh (grouped)

Run in **PowerShell**:

```
ssh-keygen -t ed25519 -f $env:USERPROFILE\.ssh\id_lab7 -C "lab_key"
```

```
ls $env:USERPROFILE\.ssh
```

```
Command Prompt
C:\Users\Lenovo>ssh-keygen -t ed25519 -f C:\Users\Lenovo\.ssh\id_lab7 -C "lab_key"
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\Lenovo\.ssh\id_lab7
Your public key has been saved in C:\Users\Lenovo\.ssh\id_lab7.pub
The key fingerprint is:
SHA256:IFwTGT+wc8Qc//891CXNvzBE6jxkwKTzJoMYmL3GTZI lab_key
The key's randomart image is:
+--[ED25519 256]--+
  ..BBo.
+ .. +Oo. .
o E...oo = + o
. B...oo = o. +
= o oSo= . .+
. + + o oo
. + o
oo
+
+----[SHA256]-----+

C:\Users\Lenovo>mkdir C:\Users\Lenovo\.ssh
A subdirectory or file C:\Users\Lenovo\.ssh already exists.

C:\Users\Lenovo>dir C:\Users\Lenovo\.ssh
Volume in drive C has no label.
Volume Serial Number is 10F6-E847

Directory of C:\Users\Lenovo\.ssh

10/01/2026 07:28 pm <DIR> .
10/01/2026 07:28 pm <DIR> ..
31/12/2025 12:50 am 419 id_ed25519
31/12/2025 12:50 am 105 id_ed25519.pub
10/01/2026 07:28 pm 399 id_lab7
10/01/2026 07:28 pm 90 id_lab7.pub
30/12/2025 02:29 pm 1,111 known_hosts
30/12/2025 02:29 pm 937 known_hosts.old
6 File(s) 3,061 bytes
2 Dir(s) 103,909,384,192 bytes free
```

Step A2: Show public key content

Run:

type %env:USERPROFILE\.ssh\id_lab7.pub

```
C:\Users\Lenovo>type C:\Users\Lenovo\.ssh\id_lab7.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICvzWj6ribTwa426rWLeKEVvOzd1AmkAQ/
```

Step A3: Clear and verify known_hosts (grouped)

Run:

type NUL > C:\Users\Lenovo\.ssh\known_hosts

type C:\Users\Lenovo\.ssh\known_hosts

```
C:\Users\Lenovo>type NUL > C:\Users\Lenovo\.ssh\known_hosts
C:\Users\Lenovo>type C:\Users\Lenovo\.ssh\known_hosts
```

Step A4: SSH to server and accept host key

Run:

ssh username@<server_ip>

- Type **yes** when prompted
- Enter your **password** (this is still password-based at this stage)

```
C:\Users\Lenovo>ssh amina@192.168.98.128
The authenticity of host '192.168.98.128 (192.168.98.128)' can't be established.
ED25519 key fingerprint is SHA256:1Q1wI2XvCxtrjZt1VMbku9cJC7+StxvDGj/RGcXsvGE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.98.128' (ED25519) to the list of known hosts.
amina@192.168.98.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jan 10 02:33:56 PM UTC 2026

System load:  0.0               Processes:            225
Usage of /:   50.5% of 14.66GB   Users logged in:     1
Memory usage: 17%              IPv4 address for ens33: 192.168.98.128
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Sat Jan 10 14:24:33 2026 from 192.168.98.1
amina@2023-bse-007:~$
```

Step A5: Show known_hosts after connection

Run:

type C:\Users\Lenovo\.ssh\known_hosts

```
C:\Users\Lenovo>type C:\Users\Lenovo\.ssh\known_hosts
192.168.98.128 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDW+9YPiiFqSnaYzyUA9S+KuV/Ic7nUcaH2c1
wAR
192.168.98.128 ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCgBLpwIQw80zwcC+k90vMed0PPaxJTEaFsn
vYD2jtmXabUCWf9qGc/g6pv0fICjLOW56qjw7UqbkMbqoiE/jO+co5uTHuifZBI73e9p9tyUM40kGwyr70PWI91
tOd6ZPb9G3/58taG5s19Fpgv6zJshih3ij0Xb7E9+a7TzTnT8eHNJtdiIj9wzid5d+k+GLgJ1ouArDlGCKqB8Q1
0o425PoncgGbQYH7fjzT79nJajNvzAaHlFgwBcPnR10qVIT+PM9UVVwsfBB5rH6aDbb82yjGb3x08pRsBLNAGm0mL
ldidG6w6AxaYk3AUxDN2UPK4NdfmHBvhmsg4SfCU+il4mORvli+W6cF1Br/7NsUpovHi6S81tKVLC/1tISWAOnZB
kBwu01odIznA1s6gS0ro2SvzY1zgfew3/LaHsZjzUwhnG4CJ41Hr6+1XvvhgbvCtGie4pE9mKkQqsdqggeDQYRxL
/q08nA/+WpnfmM=
192.168.98.128 ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoVTItbm1zdHAyNTYAAAAIbmlzdHAyNTYAAABBB
I2iyCOEt5TKVl4ldmvaCtj3zpwwtJCiTGioOss4ZPfHVe9RaqWly0x2//mHQPiTDg32xM64JCFbpdHyv1l8=

C:\Users\Lenovo>
```

🖥️ On Ubuntu Server

Step B1: Prepare .ssh and clear authorized_keys (grouped)

Run:

```
mkdir -p ~/.ssh
```

```
chmod 700 ~/.ssh
```

```
> ~/.ssh/authorized_keys
```

```
amina@2023-bse-007:~$ mkdir -p ~/.ssh
amina@2023-bse-007:~$ chmod 700 ~/.ssh
amina@2023-bse-007:~$ > ~/.ssh/authorized_keys
amina@2023-bse-007:~$
```

Step B2: Add public key & verify (grouped)

⚠️ Copy **exact public key** from id_lab7.pub (Windows)

Run on **Ubuntu**:

```
echo "ssh-ed25519 AAAA...your_full_public_key... lab_key" >> ~/.ssh/authorized_keys
```

```
chmod 600 ~/.ssh/authorized_keys
```

```
cat ~/.ssh/authorized_keys
```

```
amina@2023-bse-007:~$ echo "ssh-ed25519 AAAAC3NzaC1lZD11NTE5AAAAIDW+9YPiIFqSnaYzyUA9S+KuV/IcJnU
amina@2023-bse-007:~$ chmod 600 ~/.ssh/authorized_keys
chmod: cannot access '/home/amina/.ssh/authorized_keys': No such file or directory
amina@2023-bse-007:~$ chmod 600 ~/.ssh/authorized_keys
amina@2023-bse-007:~$ cat ~/.ssh/authorized_keys
amina@2023-bse-007:~$
```

🖥️ Test Passwordless Login (Windows)

Step C1: Normal SSH (no password)

Run on **Windows**:

```
ssh username@<server_ip>
```

```

C:\Users\Lenovo>ssh amina@192.168.98.128
amina@192.168.98.128's password:
Permission denied, please try again.
amina@192.168.98.128's password:
Permission denied, please try again.
amina@192.168.98.128's password:
amina@192.168.98.128: Permission denied (publickey,password).

C:\Users\Lenovo>_

```

Step C2: SSH with explicit identity file

Run:

ssh -i \$env:USERPROFILE\.ssh\id_lab7 amina@192.168.98.128

```

C:\Users\Lenovo>ssh -i $env:USERPROFILE\.ssh\id_lab7 amina@192.168.98.128
Warning: Identity file $env:USERPROFILE\.ssh\id_lab7 not accessible: No such file or directory.
amina@192.168.98.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jan 10 02:44:40 PM UTC 2026

System load:  0.0               Processes:            224
Usage of /:   50.5% of 14.66GB   Users logged in:     1
Memory usage: 16%              IPv4 address for ens33: 192.168.98.128
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Sat Jan 10 14:33:56 2026 from 192.168.98.1
amina@2023-bse-007:~$

```